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Temperament, Activity, and Expectations for Later Personality Development

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Temperament is assumed to be the biologically based, emotional core of personality. Adult personality is presumed to emerge developmentally from temperament. One mechanism that may link temperament to subsequent personality development involves caregiver expectancies. Stability in personality may be associated with caregiver expectancies about the meaning of temperament-based behavior. The expectancies, in combination with implicit theories of personality development, support stability and patterned change. This multimethod study examined the relations among motor activity differences, temperament, and expectations about future personality characteristics in preschool children. It was hypothesized that motor activity and temperament differences would be linked to teachers' expectations about later personality development. The hypothesis that expectations about such links would be moderated by the sex of the child was also examined. Outcomes generally corroborated hypotheses. Results are discussed in terms of personality development and age-related adaptations to social contexts.

In an apparent consensus, personality psychologists are assigning a top priority to the establishment of a language for describing the basic structure of personality (Digman, 1990; Digman & Inouye, 1986; Goldberg & Rosolack, 1994; John, 1990; McCrae & Costa, 1985). This consensus is derived from convergence in empirical work pointing toward a five-factor approach (e.g., Digman & Takemoto-Chock, 1981; Graziano, 1994; McCrae & Costa, 1987; McCrae & John, 1992; Ozer & Reise, 1994; Panter, Tanaka, & Hoyle, 1994). Some writers have claimed that the five factors have the status of "an empirical fact, like the fact that there are seven continents on earth or eight American presidents from Virginia" (McCrae & John, 1992, p. 194). Ozer and Reise (1994) suggested that just as latitude and longitude permit the precise specification of any location on earth, the five-factor system promises the hope of similarly locating personality dispositions.

Despite the apparent consensus, important "contrarian" voices are expressing reservations about the Big Five juggernaut. In a major critique, Block (1995) objected to the term *model*, suggesting the term *approach* as a more accurate substitute. He pointed to deficiencies in the five-factor approach as a general perspective on personality structure. The robustness of the five-factor model may be more apparent than real due to hidden assumptions and unrecognized or underappreciated background conditions. Block also noted the absence of an overall theoretical explanation for the five-factor approach. Similar concerns had been expressed previously by Briggs (1989), who noted that no one had an a priori theoretical reason that these five dimensions rather than some other five should emerge, nor had anyone offered a coherent, falsifiable explanation for the five-factor system.

Ozer and Reise (1994) noted that the five-factor approach should not be regarded as a theory; it is better seen as a tool for organizing phenomena to be explained by theory. Precisely how the five factors can be arranged theoretically to produce the desired quasi-cartographic coordinates, however, is not clear. One possibility is to remain as close to the data as possible and construct multidimensional, geometric categorization systems. In the abridged Big Five circumplex model (AB5C), for example, 10 two-dimensional circumplexes are produced by taking all possible pairs of the Big Five factors as coordinates (e.g., De Raad, Hendriks, & Hofstee, 1994). Geometric refinements like the AB5C can help uncover important personality phenomena (e.g., empty cell "antitypes" in natural language personality descriptions), but these approaches address only a few of the problems noted by the critics (e.g., Briggs, 1992; see also von Eye, 1990).

Another possibility, not necessarily incompatible with geometric approaches, involves a developmental analysis. The five-factor structure may be one possible outcome, among many, of the more general structural transformation of individuals over time. The five-factor dimensions may be seen as "developmental targets" or personality outcomes toward which development appears to proceed. Theoretically, the five-factor outcome is not

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necessarily inevitable, and other structural outcomes might be possible with different sociocultural constraints or developmental antecedents (Briggs, 1989; Higgins & Eccles Parsons, 1983; cf. Graziano & Ward, 1992, p. 426). In this analysis, the focus is on events and processes that might enhance or inhibit the development of the five-factor personality structure in adolescents and adults.

Both historically and theoretically, temperament has been seen as a major developmental precursor of personality structure (Rothbart & Bates, 1998; Wachs, 1994). Temperament is usually regarded as the biological foundation on which personality is based; it is the earliest expression of individual differences in personality (A. H. Buss & Plomin, 1984; Goldsmith, Losoya, Bradshaw, & Campos, 1994; Gray, 1991; Strelau, 1991). Temperament has been likened to the hard ice ball, around which the softer snowball of personality accumulates developmentally (Graziano & Waschull, 1995, pp. 240–242; Hofstee, 1991; Waddington, 1962).

Precisely how temperament differences in infants and young children are related to later personality remains unclear. One possibility is that the biological processes that produce the temperament difference (e.g., genes) also regulate through maturation behavioral expression of both temperament in early childhood and later personality (Goldsmith & Rothbart, 1991, p. 250). This approach implies stability in individual differences across the life span but does not exclude developmental change (e.g., Ahadi & Rothbart, 1994; A. H. Buss & Plomin, 1984, pp. 14–15; Caspi, 1998; Caspi & Silva, 1995; Matheny, 1989; Rothbart, 1991) nor interactions with aspects of the caregiving environment (Rothbart & Bates, 1998; Sanson & Rothbart, 1995; Wachs, 1994).

From the preceding analysis, it is possible to generate more refined alternative hypotheses about processes that might link temperament differences in young children to adult personality and to suggest how temperament may interact with aspects of the caregiving environment to produce adult personality structure. Part of the reason there is moderate stability in temperament-related behaviors across the life span may be because caregivers hold implicit beliefs about the meaning of patterns of behavior and the implications of such behavior for personality development. Using these beliefs and *implicit theories of development* (ITD), caregivers may act in ways to enhance or inhibit such development (Caspi, Bem, & Elder, 1989; Madon, Jussim, & Eccles, 1997). When a preschool child exhibits shy, withdrawn behavior during peer interaction, that behavior will elicit beliefs about the long-term consequences of such behavior, which will in turn lead to ITD-based caregiver intervention. Caregiver reaction to temperament-related behaviors should be proportional to the belief that the behavior has developmental implications. Furthermore, caregiver reactions to others who interact with the child will be influenced by ITD. Caregivers probably have more extreme reactions to an adult who makes sexual advances toward a preschooler than an adult who pushes a preschooler off a tricycle (both invidious behaviors). One reason for the difference in reaction is that the implicit developmental consequences of the former are presumably more serious than are the latter.

There are several implications of this line of theoretical development. First, temperament-based behavior in preschool children may elicit systematic reactions from caregivers based on implicit beliefs about the relation of such behavior to psychological development. The caregiver reactions are patterned vis-à-vis ITD (Goodnow & Collins, 1990; Miller, 1995; Rickard, Graziano, & Forehand, 1984). Temperament differences in young children may elicit expectations from adults not just about the schematic interconnections among contemporaneous behaviors but also about the *developmental trajectory* that a child will follow and the personality structure that the child will ultimately acquire. These expectancy-based reactions can enhance the developmental process and may even help shape the final structural outcome.

Second, it is possible that temperament-based behavior could be seen as consequential among one group of caregivers (Group A) but as inconsequential among another (Group B). If so, then the former group of caregivers may be more likely to intervene in the child's behavior than would be the latter. If the behavior is believed to be consequential and positive (e.g., cooperative, agreeable behavior), then interventions would probably promote the behavior, and we might expect greater stability in that behavior in the children in the care of Group A than in the children in the care of Group B. If the behavior is believed to be consequential and negative (e.g., attempts to resolve conflicts with physical force), then interventions would probably be designed to inhibit the behavior (Rickard et al., 1984). In this case, we should see the opposite pattern of stability for children in the care of Groups A and B: Temperament-based behavior believed to be consequential and negative should be less stable in children in the care of Group A than in those in the care of Group B. Taken together, this theorizing implies that personality development can be enhanced or inhibited by caregiver expectations and beliefs about the meaning of temperament-based behaviors and certain early life events (Caspi et al., 1989; Graziano & Eisenberg, 1997; Rickard et al., 1984). A similar analysis might be applied to therapists and other providers of health care to children (see Lengua, West, & Sandler, 1998).

A theoretical complication is that younger and older children may live in different "subcultures," with different adaptation requirements (Eccles et al., 1989; Higgins & Eccles Parsons, 1983; Lerner & Tubman, 1989). Personality attributes desirable in one childhood ecology may be less desirable in another. For example, there is evidence that classroom teachers evaluate extraverted children more positively than they evaluate introverted children in the early school years. Eliciting and shaping sociability is an important part of education in the early school years. More interesting, perhaps, is the suggestion that in later school years teachers evaluate introverted adolescents more positively than extraverted adolescents (Morris, 1979).

Difficulties in adapting to one ecology may be the result of specific individual differences required in that ecology (e.g., extraversion, conscientiousness) or the result of motive systems that generate resistance to normative pressures or authority (Graziano, Jensen-Campbell, & Hair, 1996). Low agreeableness may be a greater problem for adjustment across "subcultures" than low extraversion because agreeableness is probably related to the motivation to accommodate to various social relationships (Caspi et al., 1989; Graziano & Eisenberg, 1997; Jensen-Campbell, Graziano, & Hair, 1996). In any case, at least one five-factor dimension (extraversion) is related to patterns of life

adaptations, but reversing patterns of evaluation in younger and older persons suggests that personality interacts with the adaptive requirements of specific age-related ecologies (Graziano & Ward, 1992; West & Graziano, 1989).

Additional theoretical complications may suggest moderators for our hypotheses. Patterns of outcomes probably would be moderated by the sex of the child who emits the behavior (Feingold, 1994; Lenney, 1991). Temperament-based behaviors may be associated with different expectations for boys and girls. The same or similar behavior may be seen as a "sign" of different personality characteristics and developmental trajectories for boys and girls. For example, the same eating behavior regarded as showing a potential "lack of self-control" in a girl may be rated as no problem in a boy (Crandall, 1994). Using a behavioral confirmation paradigm, Musser and Graziano (1991) found that school-age boys and girls held the same stereotype about children younger than themselves (e.g., younger children are regarded as incompetent), but during actual face-to-face interaction boys and girls reacted differently to younger partners. Girls solicited opinions and provided nurturant support for a younger partner, but boys ignored the younger partner. The perception of incompetence apparently had different behavioral implications for school-age girls and boys. Moving past the data, we speculate that had teachers been asked to evaluate a child who had ignored a younger partner or had not shown nurturance, teachers might anticipate different implications for the girls' and boys' respective developmental trajectories. Similarly, teachers' perception of negative emotionality or sociability may be seen as having different long-term implications for boys and girls (Miller, 1995, p. 1564).

Another potential moderator is the consistency and observableness in the temperament-based behavior in the child (Kenrick & Stringfield, 1980; Plomin, DeFries, & Loehlin, 1977). One aspect of temperament in young children that may be easily observed, that shows cross-situational consistency and, in more extreme forms, that elicits interventions, is physical activity (A. H. Buss, 1991; Eaton, 1994; Eaton & Enns, 1986; Halverson & Post-Gordon, 1984). Activity has been defined in different ways, but the common element is individual differences in customary expenditure of energy through movement (Eaton, 1994). Motor activity shows sex differences (boys are more active than girls), age-related shifts (e.g., systematic net declines from ages 3 through 10), and individual differences within age groups. Individual differences emerge early in infancy, and once the neonatal period has passed, motor activity becomes associated with positive affect and social vocalizations and with parental ratings of the infant's positive emotionality. In later preschool, however, motor activity comes to be associated with problems of adjustment. Eaton and Presse (1985) found that observed activity was correlated −.43 with peer popularity in a group of preschoolers. In middle childhood, higher motor activity as measured by actometers was associated with teacher ratings of children as being less inhibited, more aggressive, more assertive, more competitive, and less cautious than lower activity peers (D. M. Buss, Block, & Block, 1980).

From the available literature, anticipating how activity differences are related conceptually to the five-factor dimensions is possible. Teacher ratings of children reported by Digman and Inouye (1986) suggest activity-related words like *energetic* and *lethargic* load on the Extraversion factor, whereas the term *restless* loads positively on Extraversion but negatively on Conscientiousness. McCrae and Costa (1987) analyzed 40 bipolar adjectives and found that higher adult activity is associated with enjoyment of social contact, dependability, achievement orientation, broad interests, and positive emotions. Eaton (1994) suggested that the developmental trajectory is from "mostly moving to mostly talking, and the residue of activity level variance in adults is seen in the context of social and verbal interaction rather than in motor behavior" (p. 182). Eaton concluded that the motorically active child becomes the extraverted adult.

This research probed the hypotheses outlined previously, using a multitrait–multimethod converging paradigm. If adult personality structure is influenced in part by caregiver expectations about the meaning of temperament-linked behavior, then assessing children's temperament to identify the most observable and reliable dimensions of temperament is necessary. If the most reliable, most conspicuous dimensions are not related to caregiver expectancies, then it is less likely that the expectancy process exerts a pervasive, systematic influence on personality development. To examine reliability and salience to caregivers, we collected converging T data (objective measures of motor activity with actometers; see Eaton, 1983) as well as R data (teacher ratings of activity and temperament). We then asked the teachers to provide ratings of their expectations for the eventual adult personality characteristics of each child, using an adaptation of Goldberg's (1992) standard markers of the five-factor dimensions.

On the basis of previous research and our hypotheses, we expected that activity differences in preschool children would be an easily observed, behavioral aspect of temperament for teachers to assess. We predicted convergence between objective measures of motor activity and teacher ratings of activity. These findings would support the claim that temperament-based behavioral differences can be detected by caregivers and their ratings converge with objective assessments, at least of the motor activity dimension.

If our hypotheses are valid, then we should also find relations between temperament-based behaviors and expectations for later personality structure. A general model summarizing our hypotheses is presented in Figure 1. More specifically, we predicted that temperament exerts its effects through direct and indirect paths. First, there is a direct path between temperament and

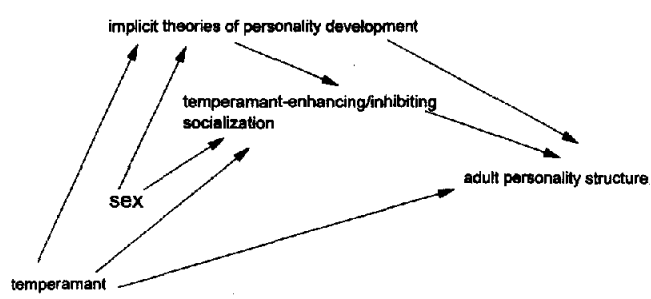

Figure 1. Implicit theories of personality development as partial mediators of the link between preschool temperament and adult personality structure.

adult personality structure because of the establishment of stable biobehavioral processes like excitation and inhibition systems. Second, temperament may elicit relatively nonreflective reactions from caregivers that enhance or inhibit temperamental tendencies (Graziano & Waschull, 1995). Third, temperamental differences may activate implicit theories of psychological development, which in turn feed into temperament-enhancing or temperament-inhibiting socialization. Sex is included in anticipation of sex-related elements in implicit theories of development and in socialization behaviors. In this model, adult personality is a developmental product of a child's temperament, in combination with caregivers' implicit theories of development and socialization behaviors.

Turning to specific temperament-related behaviors and personality characteristics, we predicted that preschool activity differences would be related to teachers' expectations for adult extraversion (positive) and conscientiousness (negative). Sociable behavior in preschool children would be related to teachers' expectations for adult extraversion and agreeableness. Emotionality in preschool children would be related (negatively) to teachers' expectations for adult emotional stability.

Method

Participants

The 69 participants (34 boys and 35 girls) were drawn from five different preschools in the Brazos Valley in central Texas. The participants ranged in age from 37 to 74 months (M = 52.46, SD = 8.07). The sample was 80% European American, 13% Mexican American, 6% African American, and 1% Asian American. Participants volunteered to participate in the study and were given parental permission for participation.

Measures

Temperament measures. To establish convergent assessment, we used two different inventories to measure each child's temperament. The first measure was the EAS Temperament Survey for Children: Teacher Ratings, which assessed the sociability, shyness, emotionality, and activity level of each child (A. H. Buss & Plomin, 1984). The second measure was the Temperament Assessment Battery for Children (TABC: Teacher Form; Martin, 1988), which assessed activity, adaptability, approach–withdrawal, emotional intensity, distractibility, and persistence.

Peer relations and interpersonal conflicts. We constructed a behavioral rating measure of quality of peer interactions. This measure consisted of 14 items that assessed same-sex and opposite-sex peer interactions. More specifically, the measure assessed whether the child interacted very little, interacted easily, was well liked, had frequent conflicts, was a leader, solved problems well, and reacted emotionally to conflict.

Expectations for adult personality structure. To measure teachers' expectations about the likely adult personality structure of the children, we used Goldberg's (1992) 100 standard markers for the five factors of personality. These served as the "developmental targets" toward which each child was presumably moving. (See *Procedure* section.) Instead of presenting them to teachers in bipolar format, we separated the poles and presented them in unipolar format. Difference scores were computed for appropriate pairs and were then used to replicate Goldberg's five-factor structure and to determine the reliability of Goldberg's markers when the targets were preschool children. (See Graziano, Jensen-Campbell, & Finch, 1997, for more details on scoring.)

Activity level. Each child's activity level was objectively measured with an actometer for 6 noncontiguous days in at least six separate sessions. The actometer methodology was developed as a tool for studying biobehavioral processes by Halverson and his colleagues (Halverson & Waldrop, 1973, 1976; Victor & Halverson, 1975) and by Eaton (1983, 1994; Eaton & Enns, 1986). The actometer device consists of two modified self-winding mechanical watches that measure separately vertical and horizontal movement. These devices were placed in special pouches attached to the child's back. Previous research has found that accurately calibrated actometers yield reliable measures of children's activity levels (Halverson & Post-Gordon, 1984).

Procedure

The experiment took place in two phases. During the first phase, each child's activity level was assessed using the actometers. To receive accurate measurements, actometers were calibrated once a week by first setting the modified watches to 12:00. We then attached the devices to a door and swung the door for at least 15 min. All measurements were then read to make sure that they were within a few seconds of each other. Any actometers that did not calibrate properly were removed.

Once the watches were calibrated, they were fastened with Velcro to a cookie sheet for transportation. This was done to make sure all watches were at the same setting at the beginning of each session. Groups of 5–7 children were tested at a time. Children were told that we were interested in how much they liked to play. We then told them that we wanted to place the actometer on their back and it would help us determine how much they liked to play. Each actometer was placed squarely between the child's shoulder blades to prevent the child from touching the device. In addition, we put an actometer on each child's back for several days so he or she could get used to it before we took our actual measurements.

After the actometer was placed on the child's back, we started the stopwatch that corresponded to that actometer. The stopwatches allowed us to obtain precise measurements of the duration that each actometer was on the child's back. The actometers were left on each child for an average of 32 min per session. Each child wore the watch for three morning and three afternoon sessions in 6 noncontiguous days to control for time-of-day effects. In addition, actometers were counterbalanced across experimental sessions. All experimental sessions consisted of outdoor play, to allow for more variability in activity (vs. the constraints imposed by indoor play; see Martin, Wisenbaker, & Huttunen, 1994, p. 163, for rationale). When the session was completed, the actometers were then taken off the child and placed back on the cookie sheet and then taken to a safe place for readings.

During the second phase, teachers were asked to provide their ratings of each child's temperament using paper-and-pencil versions of the teacher TABC and EAS. The TABC consisted of 48 items that asked teachers to rate how often behaviors occurred using Likertized scales ranging from 1 (*hardly ever*) to 7 (*almost always*). Teachers were asked to make this judgment based on how they thought the child compares with other children around the same age and based on the child's behavior in the last 3 months. The EAS also asked teachers to rate the behaviors of children on a scale ranging from 1 (*not characteristic or typical of the child*) to 5 (*very characteristic or typical of the child*). If the teacher did not have experience observing the child, they were asked to mark *not observed.* (No teacher marked *not observed.*)

Teachers were then asked to rate each child's quality of peer interactions using the measure described previously. Teachers rated behaviors associated with peer interactions on a Likertized scale ranging from 1 (*not at all accurate*) to 5 (*very accurate*).

Finally, the teachers completed Goldberg's (1992) 100 markers with children as the target of rating. Teachers were told that the words were common human traits used to describe adults. We then told them that we realized that many words did not accurately describe children of preschool age. People often have hunches, however, about how children will turn out. We then told them that we would like them to use the list

to describe how they thought the child might or might not be in the future, at approximately 20 years of age. For each trait, the teacher was asked to rate the child using a Likertized scale ranging from 1 (*not at all accurate*) to 5 (*very accurate*).

Results

Measures of Internal Consistency and Convergence

We began by probing the internal consistencies of our measures and then the convergence between ratings of the TABC and the EAS. First, the internal consistencies were computed for each standardized temperament measure using Cronbach's alpha. The alphas for the six TABC scales of Activity, Adaptability, Approach–Withdrawal, Intensity, Distractibility, and Persistence were .78, .60, .76, .64, .57, and .65, respectively. The alphas for the four EAS scales of Shyness, Sociability, Emotionality, and Activity were .57, .65, .91, and .73, respectively.

Teachers' ratings of TABC activity converged with ratings of EAS Activity, $r(67) = .71$, $p < .01$. Shyness ($r = -.75$) and Sociability ($r = .52$) also converged with TABC Approach–Withdrawal ratings. Moreover, EAS Emotionality converged with TABC Emotional Intensity ($r = .58$). Intercorrelations among temperament measures are reported in Table 1.

Next, internal consistencies were computed for each Goldberg (1992) personality dimension using Cronbach's alpha. The alphas for Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Intellect were .91, .94, .93, .88, and .91, respectively. The peer interactions measure had an overall alpha of .64 and alphas of .51 and .55 for same-sex and opposite-sex peer relations, respectively. Because of these low reliabilities, the scale was disaggregated into three subscales: peer interactions (interacts very little; interacts very easily), peer conflicts (has frequent conflicts; solves conflicts well; reacts emotionally to conflicts), and likability by peers (seems to be well liked; is frequently a leader). The alphas for the subscales were .64, .83, and .73, respectively. All negative items (e.g., interacts very little) were reverse coded to create summed composite scores.

Finally, the internal consistencies of the actometer data were computed using Cronbach's alpha. The alphas for vertical and horizontal movement across the six measurement sessions were .77 and .66, respectively. Moreover, the two measures were significantly correlated with one another, $r(67) = .69$, $p < .01$. Due to the lower reliability of horizontal movement and the lower relation between the two measures than in previous studies (cf. Halverson & Waldrop, 1973; $rs > .87$), the two measures of movement were kept as separate indexes of activity level.

Activity Levels

Preliminary analyses. The measures of vertical and horizontal movement were computed first by taking each session's measurements and converting the measures into "minutes." (Actometer readings included both "minute" and "second" readings.) Next, we aggregated the six session readings and divided by the actual time the actometers were worn as recorded by the stopwatches. (The stopwatch time was also converted into minute readings.)

Next, we correlated vertical and horizontal movements with the sex of the child. There was no evidence of a relation between the sex of the child and total summed objective vertical and horizontal movement ($rs = .07$ and $-.09$, respectively). For vertical movement, there was no evidence that boys ($M = .09$, $SD = .03$) moved more than did girls ($M = .09$, $SD = .05$), $t(67) = -.60$, *ns*. For horizontal movement, there was no evidence that boys ($M = .25$, $SD = .08$) moved more than did girls ($M = .23$, $SD = .11$), $t(67) = .74$, *ns*.

Finally, we correlated vertical and horizontal movement with the age of the child. There was no evidence of a relation between age (in months) and vertical and horizontal movement ($rs = .15$ and $.19$, respectively). Moreover, age distributions for boys and girls did not differ (Ms = 52.79 and 52.14, SDs = 8.91 and 7.29, respectively), $t(67) = .33$, *ns*.

Tests of activity hypotheses. We predicted that objectively measured activity levels (i.e., actometer data) were related to teachers' perceptions of preschoolers' temperament. More specifically, we expected actometer readings to be related to contemporaneous measures of temperament, which are related to activity and extraversion (i.e., TABC Activity, TABC Approach–Withdrawal, EAS Shyness, EAS Sociability, and EAS Activity).

Table 1
Intercorrelations Among Temperament Measures

Measure	1	2	3	4	5	6	7	8	9	10
TABC Measures										
1. Activity	—									
2. Adaptability	−.19	—								
3. Approach–Withdrawal	.24*	.44**	—							
4. Emotional Intensity	.51**	−.12	.46**	—						
5. Distractability	.70**	−.25**	.19	.59**	—					
6. Persistence	−.54**	.49**	.27**	−.28*	−.59**	—				
EAS Measures										
7. Activity	.71**	−.05	.33**	.53**	.46**	−.36**	—			
8. Shyness	−.28*	−.25*	−.75**	−.46**	−.26*	−.05	−.37**	—		
9. Sociability	.49**	.21†	.52**	.55**	.38**	−.14	.58**	−.69**	—	
10. Emotionality	.35**	−.19	.26*	.58**	.50**	−.31**	.30**	−.18	.20	—

Note. TABC = Temperature Assessment Battery for Children; EAS = A. H. Buss and Plomin's (1984) EAS Temperature Survey.
† $p < .10$ (marginally significant). * $p < .05$. ** $p < .01$.

Table 2
Intercorrelations Among Actometer Readings and Teacher Ratings of Temperament and Dimensions of Personality

	Actometer readings					
	Overall		Boys		Girls	
Measures	Vertical	Horizontal	Vertical	Horizontal	Vertical	Horizontal
EAS measures						
Activity	.12	.13	.30†	.16	.07	.08
Shyness	−.26*	−.27*	−.41*	−.25	−.17	−.29†
Sociability	.18	.21†	.32†	.24	.13	.17
Emotionality	−.26*	−.18	−.05	−.07	−.39*	−.24
TABC measures						
Activity	.04	−.05	.35*	.00	−.07	−.12
Adaptability	.02	−.02	−.01	.07	.04	−.10
Approach–Withdrawal	.18	.17	.42**	.31†	.02	.07
Emotional Intensity	.04	.13	.27	.21	−.05	.08
Distractibility	−.01	−.13	.14	−.14	−.09	−.12
Persistence	.08	.06	.08	.07	.07	.07
Goldberg markers						
Extraversion	.22†	.07	.51**	.25	.01	−.09
Agreeableness	.25*	.13	.09	.04	.33*	.21
Conscientiousness	.24*	.10	.14	.07	.35*	.14
Emotional Stability	.10	.01	−.13	−.10	.34*	.11
Intellect	.32**	.09	.25	.15	.42**	.06

Note. EAS = A. H. Buss and Plomin's (1984) EAS Temperament Survey; TABC = Temperament Assessment Battery for Children.
† $p < .10$ (marginally significant). * $p < .05$. ** $p < .01$.

Within this framework, we also examined the possibility that the sex of the child would affect teachers' perceptions of them. Operationally, vertical and horizontal movement were treated as continuous variables and centered to avoid problems with unstandardized solutions (Aiken & West, 1991). To analyze teachers' evaluations of children, iterative sets of analyses for vertical and horizontal movements were conducted using a 2 (activity level: horizontal or vertical movement) × 2 (sex of child) factorial design.[1] The dependent measures were the teachers' ratings of the child's temperament and personality. Because of space limitations, only significant findings will be reported.

First, as mentioned above, we predicted that objectively measured activity (i.e., actometer readings) would be related to teachers' contemporaneous ratings of temperament. For the TABC Activity subscale, there was a significant main effect for sex of the child, $F(1, 65) = 7.16$, $p < .01$. Boys were rated as having lower mean activity levels ($M = 28.15$, $SD = 6.98$) than girls ($M = 32.83$, $SD = 9.86$). There was also a marginal Sex of Child × Vertical Movement interaction, $F(1, 65) = 2.61$, $p = .11$. For boys, vertical movement was significantly related to reports of activity, $r(32) = .35$, $p < .04$ (see Table 2). For girls, there was no evidence that vertical movement was related to reports of activity, $r(33) = -.07$, *ns*. For the TABC Approach–Withdrawal subscale, there was a vertical movement main effect, $F(1, 65) = 4.90$, $p < .03$. There was also a significant Sex of Child × Vertical Movement interaction, $F(1, 65) = 4.28$, $p < .04$. For boys, approach–withdrawal was significantly related to vertical movement, $r(32) = .42$, $p < .05$. For girls, there was no evidence that vertical movement was related to approach–withdrawal behavior, $r(33) = .02$, *ns*. There was no evidence of any relation between the subscales of the TABC and horizontal movement.

For the EAS Shyness subscales, there was a main effect for vertical movement, $F(1, 65) = 7.20$, $p < .01$. Vertical movement was negatively correlated with ratings of Shyness, $r(67) = -.26$, $p < .03$. There was also a horizontal movement main effect, $F(1, 65) = 4.89$, $p < .03$, $r(67) = -.27$, $p < .02$. For the EAS Sociability subscale, there was a main effect for vertical movement, $F(1, 65) = 3.52$, $p < .06$. For boys, there was a relation between Sociability and vertical movement, $r(32) = .32$, $p < .06$. For girls, there was no evidence of a relation between Sociability and vertical movement, $r(33) = .13$, *ns*. There was also a marginal main effect for horizontal movement, $F(1, 65) = 2.84$, $p = .09$, $r(67) = .21$, $p = .08$. For EAS Emotionality, there was a marginally significant main effect for vertical movement, $F(1, 65) = 2.66$, $p = .10$, $r(67) = -.26$, $p < .03$.

Next, we predicted that objectively measured activity levels would be related to implicit theories of personality development. We used the dimensions of the five-factor approach as the developmental targets of teachers' expectancies. For expected adult

[1] Analyses with age as an independent variable were also conducted. Age was treated as a continuous variable and centered to avoid problems with unstandardized solutions. Age was significantly related to Adaptability, Approach–Withdrawal, Shyness, and Sociability ($rs = .27, .39, -.25, .31$, respectively, $ps < .05$). Activity, however, still contributed uniquely to predicting temperament and ITDs. Moreover, age did not systematically moderate activity effects. Given our small sample size, we collapsed across age differences for tests of our hypotheses.

Extraversion, there was a main effect for vertical movement, $F(1, 65) = 7.91$, $p < .01$. There was also a Sex of Child × Vertical Movement interaction, $F(1, 65) = 7.38$, $p < .01$. For boys, expectation for adult Extraversion was significantly related to vertical movement, $r(32) = .51$, $p < .05$. For girls, there was no evidence of a relation between expectations for Extraversion and vertical movement, $r(33) = .01$, *ns*. For expected adult Agreeableness, there was a marginal main effect for vertical movement, $F(1, 65) = 2.50$, $p = .11$, $r(67) = .25$, $p < .04$. For boys, there was no evidence of a relation between expected adult Agreeableness and vertical movement, $r(32) = .09$, *ns*. For girls, there was a relation between expected adult Agreeableness and vertical movement, $r(33) = .33$, $p < .05$. For expected adult Conscientiousness, there was also a marginally significant main effect for vertical movement, $F(1, 65) = 3.04$, $p = .08$, $r(67) = .24$, $p < .04$. For boys, there was no evidence of a relation between expected adult Conscientiousness and vertical movement, $r(32) = .14$, *ns*. For girls, there was a significant relation between expected adult Conscientiousness and vertical movement, $r(33) = .35$, $p < .05$. For expected adult Openness to Experience, there was a main effect for vertical movement, $F(1, 65) = 6.74$, $p < .01$, $r(67) = .32$, $p < .01$. The absolute magnitude of the association was more than twice as large for girls, $r(33) = .42$, $p < .01$, than for boys, $r(32) = .25$, *ns*. There was no evidence of any relation between the five factors of personality and horizontal movement.

Temperament and the Five-Factor Structure of Personality

Next, we probed the hypothesis that perceptions of preschool temperament would predict teachers' implicit personality theories of what children will be like as adults. To analyze teachers' implicit theories, we conducted iterative sets of general linear model equations. In other words, the TABC and the EAS subscales were examined in separate sets of analyses to probe the relation between temperament and the five factors of personality.

First we expected that temperament measures related to activity (i.e., TABC Activity, TABC Approach–Withdrawal, EAS Shyness, EAS Sociability, and EAS Activity) would be related to implicit theories of Extraversion development. TABC Activity ($b = .41$) and Approach–Withdrawal ($b = .77$) significantly predicted expectations of Extraversion in adulthood, $t(62) = 2.05$ and 3.14, ps $< .04$, respectively. EAS Shyness ($b = -2.02$) was also negatively related to expectations of adult Extraversion, $t(64) = -4.21$, $p < .001$. (See Table 3 for zero-order correlations.)

Next, we expected that temperament measures related to adaptability and effortful control would be linked to teachers' implicit theory of Agreeableness and Conscientiousness development (i.e., TABC Adaptability, TABC lack of Emotional Intensity, EAS lack of Emotionality, and TABC Persistence; Ahadi & Rothbart, 1994; Martin et al., 1994). TABC Activity level ($b = -.41$) and Emotional Intensity ($b = -.52$) were marginally negatively related to expectations of Agreeableness, $t(62) = -1.69$ and -1.92, ps $< .10$, respectively. Moreover, TABC Adaptability ($b = .81$) positively predicted expectations of Agreeableness, $t(62) = 2.66$, $p < .01$. EAS Emotionality ($b = -1.19$) and Activity ($b = -1.91$) were also negatively related to expectations of adult Agreeableness, $t(64) = -4.54$ and -4.60, respectively. Moreover, EAS Shyness ($b = -.87$) was marginally and negatively related to Agreeableness, $t(64) = -1.74$, $p = .08$. TABC Adaptability ($b = .59$) and Persistence ($b = .77$) predicted levels of Conscientiousness, $t(62) = 2.50$ and 2.96, ps $< .02$, respectively. EAS Emotionality ($b = -.77$) was negatively related to expectations of adult Conscientiousness, $t(64) = -2.96$, $p < .01$.

We then predicted that temperament measures of emotionality (i.e., TABC Emotional Intensity and EAS Emotionality) would be related to expectations of adult Emotional Stability. Recall that Emotional Stability is anchored positively (Goldberg, 1992). That is, higher scores reflect more Emotional Stability rather than more Neuroticism, as is typical in other five-factor scale solutions. TABC Adaptability ($b = .71$) and Persistence ($b = .54$) positively predicted Emotional Stability, $t(62) = 3.06$ and 2.09, respectively (ps $< .05$). TABC Approach–Withdrawal ($b = -.65$) was negatively related to Emotional Stability, $t(62) = -2.90$, $p < .01$. Moreover, TABC Activity level ($b = .31$) was marginally related to expectations of Emotional Stability, $t(62) = 1.72$, $p = .09$. EAS Emotionality ($b = -1.32$) was negatively related to adult Emotional Stability, $t(64) = -6.44$, $p < .001$.

Information about the developmental origins of the personality dimension of Intellect/Openness to Experience is limited (McCrae & Costa, 1997). Nevertheless, we explored the possible links between preschool measures of temperament and implicit theories of the development of this dimension. TABC Adaptability ($b = .63$) and Approach–Withdrawal ($b = .48$) positively predicted Openness to Experience, $t(62) = 2.53$ and 1.99, respectively. EAS Shyness ($b = -1.81$) was negatively related to Intellect, $t(64) = -3.83$, $p < .01$. EAS Emotionality ($b = -.41$) was also marginally and negatively related to expectations of adult Intellect, $t(64) = -1.68$, $p = .09$.

Supplementary Analyses

Next, we probed the links among temperaments, expected adult personality, and quality of peer interactions. Iterative sets of regression analyses were again conducted. We began by probing the relation between temperament and quality of peer interactions. We then used the expected five factors to predict teachers' perceptions of quality peer interactions.

Temperament ratings. First, we probed the relation between preschool temperament and quality of peer interactions. TABC Adaptability ($b = .23$) was predictive of the quality of peer interactions, $t(62) = 3.80$, $p < .01$. TABC Activity level ($b = .09$) was also marginally related to interactions, $t(62) = 1.86$, $p = .07$. EAS Shyness ($b = -.27$) and Emotionality ($b = -.12$) were negatively related to quality of peer interactions, $t(64) = -2.32$ and -2.03, respectively, ps $< .05$. Moreover, EAS Sociability ($b = .37$) also significantly predicted quality interactions, $t(64) = 2.33$, $p < .02$.

Next we probed the relation between preschool temperament and patterns in handling conflicts with peers. TABC Emotional Intensity ($b = -.22$) was negatively related to handling conflicts well, $t(62) = -2.57$, $p < .01$. Adaptability ($b = .17$) also marginally predicted handling conflicts well, $t(62) = 1.78$, $p = .08$. EAS Sociability ($b = .37$) and Emotionality ($b = -.66$)

Table 3
Intercorrelations Among Teacher Ratings of Temperament and Dimensions of Personality

Measure	Goldberg markers				
	Extraversion	Agreeableness	Conscientiousness	Emotional stability	Intellect
EAS measures					
Activity	.31**	−.48**	−.26*	−.14	−.10
Shyness	−.57**	−.06	.01	.23†	−.39**
Sociability	.39**	−.08	−.09	−.16	.13
Emotionality	.09	−.52**	−.39**	−.64**	−.17
TABC measures					
Activity	.40**	−.43**	−.35**	−.26*	.03
Adaptability	.26*	.48**	.48**	.36**	.50**
Approach–Withdrawal	.56**	.09	.13	−.16	.48**
Emotional Intensity	.30*	−.38**	−.23†	−.35**	.08
Distractibility	.31*	−.39**	−.46**	−.42**	−.05
Persistence	.01	.45**	.60**	.39**	.29**

Note. EAS = A. H. Buss and Plomin's (1984) EAS Temperament Survey; TABC = Temperament Assessment Battery for Children.
† $p < .10$ (marginally significant). * $p < .05$. ** $p < .01$.

also predicted the preschoolers' ability to handle conflicts well, $t(64) = 2.02, -9.36$, $ps < .05$, respectively.

Finally, we probed the link between temperament and the likability of the preschool children. EAS Activity level ($b = -.30$) was negatively related to likability by peers, $t(64) = -2.62$, $p < .01$. There was no evidence that any TABC component predicted likability by classmates.

Five structural dimensions of personality. We then probed the relation between expectations about personality development and preschool measures of peer interactions. There are several ways to examine this idea. To minimize the possible number of analyses, we used expectations for adult personality for each of the five factors as predictors of the criterion of peer relations. This method allowed us to examine how peer relations are related to each of the five factor dimensions after removing possible overlap in prediction variance among the five personality dimensions. Quality of peer interactions was significantly predicted by Extraversion ($b = .12$), $t(63) = 3.77$, $p < .01$. Moreover, handling conflicts well was significantly predicted by Agreeableness ($b = .10$) and by Emotional Stability ($b = .19$), $t(63) = 2.51$ and 3.98, respectively, $ps < .01$. Agreeableness ($b = .08$) also significantly predicted likability by peers, $t(63) = 2.04$, $p < .05$.

Discussion

Temperament is commonly described as the biologically based, emotional core of personality. The structure of adult personality is presumed to emerge developmentally from temperament. As mentioned previously, temperament has been likened to a hard ice ball, around which the softer snowball of personality accumulates developmentally. Metaphors of this sort can have heuristic value, but when they do not specify mechanisms of transformation, they fall short as tools for scientific analysis. It is one thing to paint a colorful metaphor for personality development and another to specify transformational mechanisms (D. M. Buss, 1990, 1995; Halverson, 1988; Lewis, 1997; Tooby & Cosmides, 1990) that yield adult personality structure.

Outcomes of the present research suggest that one set of mechanisms that may link early temperamental differences to later personality structure involves expectations of socializing agents. Teachers not only observe temperament-related individual differences in preschool children, but they also anticipate developmental consequences for these individual differences. Preschool teachers expect adult extraversion from children who are not shy, have high-quality peer interactions, and (for boys) are physically active. Preschool teachers expect adult Agreeableness from children who are relatively adaptable, handle interpersonal conflict well, and are liked by peers. In light of recent empirical work on the five-factor approach, these expectations are not wildly off the mark (Graziano et al., 1996; Graziano, Hair, & Finch, 1997; Graziano, Jensen-Campbell, & Finch, 1997; Jensen-Campbell et al., 1996). Taken together, our outcomes imply that adult caregivers expect that preschool behaviors are linked to adult personality structure and that agreeableness and extraversion are developmentally stable aspects of persons. A further implication, albeit a speculative one, is that personality development can be enhanced or inhibited by caregiver expectations and beliefs about the meaning of temperament-based behaviors and certain early life events (Caspi, 1998; Caspi et al., 1989; Graziano & Eisenberg, 1997; Rickard et al., 1984).

Conceptualizing preschool teachers' expectations about adult personality as a single, monolithic mechanism linking early temperament differences to later personality structure is probably not accurate. Expectations of the sort we collected are probably part of a social–cognitive matrix that includes ITDs. ITDs would contain assumptions about the relation between a specific behavior and some corresponding latent trait, but more important, they would contain implicit beliefs about the way such latent traits and behavior combine with life experience over time. Little is known about ITDs, but presumably they would be more developmentally dynamic versions of implicit theories of mind (e.g., Bartsch & Wellman, 1995). It is also reasonable to assume that ITDs contain action implications for socializing

agents (Rickard et al., 1984). If a preschool teacher's ITD contains the belief that harsh parental punishment of children leads to poor school adjustment (Weiss, Dodge, Bates, & Pettit, 1992), then the teacher may take action against parents who she or he believes are using harsh punishment.

One issue in critical need of future theoretical development involves the interpersonal, socially shared nature of ITDs among different socializing agents (e.g., mothers vs. fathers, parents vs. teachers, majority vs. minority teachers). The issue involves more than the problem of convergence in assessment across raters and instruments (see Lengua et al., 1998). There may be higher level theoretical consequences of socially shared ITDs for personality development. We must be speculative here because there is no organized empirical literature. For example, ITDs may be shared within families, and even within large social groups, and passed from generation to generation. There is evidence that distinctive patterns of affective expression can be passed from generation to generation across hundreds of years within localized groups (e.g., Collett, 1984). Similar intergenerational transmission may occur for social–cognitive beliefs about development. For another example, when parents do not share ITDs, a child may receive inconsistent parental socialization, which in turn may produce a less interconnected personality structure (e.g., Graziano & Bryant, 1998; Halverson & Wampler, 1997; Kernis & Waschull, 1995; Martin & Halverson, 1991) and may make persons more difficult for other persons to judge compared with those with more consistent parental socialization (e.g., Colvin, 1993; Davis & Kraus, 1997; Funder & West, 1993). For another example at the interpersonal level, a parent who does not hold the same ITD as his child's preschool teacher about punishment may react with indignation at the apparently inappropriate intrusion into the parent–child relationship.

If expectations and ITDs are part of a complex matrix that contributes to the developmental transformation of personality, then it might be argued that individual differences in children—and subsequent correlations with later personality structure—are illusions created by the expectations of socializing agents. Given the potentially powerful impact of ecological demands (e.g., Eccles et al., 1989; Higgins & Eccles Parsons, 1983), situated, age-related social norms can create regularities in behavior that might be perceived as stable personality differences by observers (e.g., Lewis, 1997). This interpretation, at least as a general account of temperament-related aspects of personality, is too radical for several reasons. First, the "flow of influence" in socialization travels not only from caregiver to child, but also from child to caregiver (Graziano & Waschull, 1995; Halverson & Wampler, 1997; Plomin, DeFries, & Loehlin, 1977; Scarr & McCartney, 1983). That is, temperament may provide the raw material out of which expectancy-based socialization may emerge. Temperament-based behaviors may elicit patterned caregiver socialization. The child may respond in ways consistent with the socialization or may actively resist (e.g., Rothbart & Bates, 1998; Wachs, 1994, p. 217). Resistance may be less common in younger children than in older individuals, however, at least for behaviors tied to established self-concepts. Promising research is now emerging on processes that moderate the impact of expectancies (Madon et al., 1997; Musser & Graziano, 1991; Olson, Roese, & Zanna, 1996; Swann & Ely, 1984).

A second argument against the radical, idiosyncratic fabrication view comes from previous research showing patterns in expectancies among caregivers. Rickard et al. (1984) examined variables related to parental beliefs about child development norms and found that inaccurate and unrealistic parental expectations were linked to child-control problems. Unrealistic expectations are more common among parents of firstborns, young parents, less well-educated parents, and fathers. Beyond these group trends, Rickard et al. showed that the parents least able to control their children at home were the ones with the least accurate knowledge about child development norms. The Rickard et al. results suggest that some ITDs are more accurate than others and have practical use as tools of socialization. Given these relations, it might be possible to develop educational interventions to help caregivers develop more accurate ITDs about children.

Into this hopeful note we introduce a theoretical complication. Implicit in the previous discussion was the notion of "accuracy." Because we have such limited knowledge about the reliability and cross-group generality of effects in personality development and their link to socialization, it is difficult to define definitively what is accurate knowledge. Even caregiving involving harsh physical punishment bordering on child abuse does not invariably produce psychopathology in all children (e.g., Weiss et al., 1992). In the past, most researchers have dealt with this accuracy problem largely by working with the proxy of interparental agreement (Halverson & Wampler, 1997). Disagreement between parents implies that somebody is probably not accurate, but interparental agreement, per se, does not guarantee accuracy because both parents could be completely inaccurate. In the future, we will need more sophisticated theoretical development linking intercaregiver agreement and accuracy.

The radical illusion view is implausible, but lacking established empirical relations, ruling it out definitively at this time is impossible. Without denying the potency of either ecological demands or temperaments, however, we believe that our data are best described in terms of an interactive process, in which characteristics of the child in combination with the demands of the specific ecology and expectations of the caregiver elicit differences in socialization (see Halverson & Wampler, 1997; Ickes, 1982, pp. 327–328). Our data on expectations for developmental outcomes associated with preschool activity differences in boys and girls seem consistent with an interactive perspective.

Outcomes from the present research suggest that considering the structural approach to personality from a developmental perspective may be profitable. The five-factor structure may be one possible outcome, among many, of the more general structural transformation of individuals over time. The five-factor dimensions may be seen as developmental targets or personality outcomes toward which development appears to proceed. Theoretically, the five-factor outcome is not necessarily inevitable, and other structural outcomes might be possible with different sociocultural constraints or developmental antecedents (Briggs, 1989; Higgins & Eccles Parsons, 1983; cf. Hofstee, 1991, p. 180). In this analysis, the focus is not on the pervasiveness or robustness of the five-factor structure but on processes that might enhance or inhibit the development of personality struc-

ture, which is itself one part of the transformation of individuals over time.

References

Ahadi, S. A., & Rothbart, M. K. (1994). Temperament, development, and the Big Five. In C. F. Halverson, G. A. Kohnstamm, & R. P. Martin (Eds.), *The developing structure of temperament and personality from infancy to adulthood* (pp. 189–208). Hillsdale, NJ: Erlbaum.

Aiken, L. S., & West, S. G. (1991). *Multiple regression: Testing and interpreting interactions.* Newbury Park, CA: Sage.

Bartsch, K., & Wellman, H. M. (1995). *Children talk about the mind.* New York: Oxford University Press.

Block, J. (1995). A contrarian view of the five-factor approach to personality description. *Psychological Bulletin, 117,* 187–215.

Briggs, S. R. (1989). The optimal level of measurement for personality constructs. In D. M. Buss & N. Cantor (Eds.), *Personality psychology: Recent trends and emerging directions* (pp. 246–260). New York: Springer.

Briggs, S. R. (1992). Assessing the five-factor model of personality description. *Journal of Personality, 60,* 253–294.

Buss, A. H. (1991). The EAS theory of temperament. In J. Strelau & A. Angleitner (Eds.), *Exploration in temperament: International perspectives on theory and measurement* (pp. 43–60). New York: Plenum.

Buss, A. H., & Plomin, R. (1984). *Temperament: Early developing personality traits.* Hillsdale, NJ: Erlbaum.

Buss, D. M. (1990). Toward a biologically informed psychology of personality. *Journal of Personality, 58,* 1–16.

Buss, D. M. (1995). Evolutionary psychology: A new paradigm for psychological science. *Psychological Inquiry, 6,* 1–30.

Buss, D. M., Block, J. H., & Block, J. (1980). Preschool activity level: Personality correlates and developmental implications. *Child Development, 51,* 401–408.

Caspi, A. (1998). Personality development across the life course. In W. Damon (Series Ed.) & N. Eisenberg (Vol. Ed.), *Handbook of child psychology: Vol. 3. Social, emotional, and personality development* (5th ed., pp. 311–387). New York: Wiley.

Caspi, A., Bem, D., & Elder, G. H., Jr. (1989). Continuities and consequences of interaction styles across the life course. *Journal of Personality, 57,* 375–406.

Caspi, A., & Silva, P. A. (1995). Temperament qualities at age three predict personality traits in young adulthood: Longitudinal evidence from a birth cohort. *Child Development, 66,* 486–498.

Collett, P. (1984). History and the study of expressive action. In K. Gergen & M. Gergen (Eds.), *Historical social psychology* (pp. 371–396). Hillsdale, NJ: Erlbaum.

Colvin, C. R. (1993). "Judgable" people: Personality, behavior, and competing explanations. *Journal of Personality and Social Psychology, 64,* 861–873.

Crandall, C. S. (1994). Prejudice against fat people: Ideology and self-interest. *Journal of Personality and Social Psychology, 66,* 882–894.

Davis, M. H., & Kraus, L. A. (1997). Personality and empathic accuracy. In W. Ickes (Ed.), *Empathic accuracy* (pp. 144–168). New York: Guilford Press.

De Raad, B., Hendriks, A. A. J., & Hofstee, W. K. B. (1994). The Big Five: A tip of the iceberg of individual differences. In C. F. Halverson, G. A. Kohnstamm, & R. Martin (Eds.), *The developing structure of temperament and personality from infancy to adulthood* (pp. 91–110). Hillsdale, NJ: Erlbaum.

Digman, J. H. (1990). Personality structure: The emergence of the five-factor model. *Annual Review of Psychology, 41,* 417–440.

Digman, J. M., & Inouye, J. (1986). Further specification of the five robust factors of personality. *Journal of Personality and Social Psychology, 50,* 116–123.

Digman, J. M., & Takemoto-Chock, N. K. (1981). Factors in the natural language of personality: Re-analysis and comparison of six major studies. *Multivariate Behavioral Research, 16,* 149–170.

Eaton, W. O. (1983). Measuring activity level with actometers: Reliability, validity, and arm length. *Child Development, 54,* 720–726.

Eaton, W. O. (1994). Temperament, development, and the five-factor model: Lessons from activity level. In C. F. Halverson, D. Kohnstamm, & R. Martin (Eds.), *The developing structure of temperament and personality from infancy to adulthood* (pp. 173–188). Hillsdale, NJ: Erlbaum.

Eaton, W. O., & Enns, L. R. (1986). Sex differences in human motor activity. *Psychological Bulletin, 100,* 19–28.

Eaton, W. O., & Presse, M. C. (1985). *Impact, behavioral quality, and popularity in preschool children.* Unpublished manuscript, Department of Psychology, University of Manitoba, Winnipeg, Canada.

Eccles, J., Wigfield, A., Flanagan, C., Miller, C., Reuman, D., & Yee, D. (1989). Self-concepts, domain values, and self-esteem: Relations and changes in early adolescence. *Journal of Personality, 57,* 283–310.

Feingold, A. (1994). Gender differences in personality: A meta-analysis. *Psychological Bulletin, 116,* 429–456.

Funder, D. C., & West, S. G. (1993). Consensus, self–other agreement, and accuracy in personality judgment: An introduction. *Journal of Personality, 61,* 457–476.

Goldberg, L. R. (1992). The development of markers for the Big-Five factor structure. *Psychological Assessment, 4,* 26–42.

Goldberg, L. R., & Rosolack, T. K. (1994). The Big Five factor structure as an integrative framework: An empirical comparison with Eysenck's P-E-N model. In C. F. Halverson, G. A. Kohnstamm, & R. Martin (Eds.), *The developing structure of temperament and personality from infancy to adulthood.* (pp. 7–36). Hillsdale, NJ: Erlbaum.

Goldsmith, H. H., Losoya, S. H., Bradshaw, D. L., & Campos, J. J. (1994). Genetics of personality: A twin study of the five-factor model and parent–offspring analyses. In C. F. Halverson, D. Kohnstamm, & R. Martin (Eds.), *The developing structure of temperament and personality from infancy to adulthood* (pp. 241–266). Hillsdale, NJ: Erlbaum.

Goldsmith, H. H., & Rothbart, M. K. (1991). Contemporary instruments for assessing early temperament by questionnaire and the laboratory. In J. Strelau & A. Angleitner (Eds.), *Exploration in temperament: International perspectives on theory and measurement* (pp. 249–272). New York: Plenum.

Goodnow, J. J., & Collins, W. A. (1990). *Development according to parents: The nature, sources, and consequences of parents' ideas.* London: Erlbaum.

Gray, J. A. (1991). The neuropsychology of temperament. In J. Strelau & A. Angleitner (Eds.), *Exploration in temperament: International perspectives on theory and measurement* (pp. 105–128). New York: Plenum.

Graziano, W. G. (1994). The development of agreeableness as a dimension of personality. In C. F. Halverson, D. Kohnstamm, & R. Martin (Eds.), *The developing structure of temperament and personality from infancy to adulthood* (pp. 339–354). Hillsdale, NJ: Erlbaum.

Graziano, W. G., & Bryant, W. (1998). Self-monitoring and the self-attribution of positive emotion. *Journal of Personality and Social Psychology, 74,* 250–261.

Graziano, W. G., & Eisenberg, N. (1997). Agreeableness: A dimension of personality. In R. Hogan, J. Johnson, & S. Briggs (Eds.), *Handbook of personality psychology* (pp. 795–824). Orlando, FL: Academic Press.

Graziano, W. G., Hair, E. C., & Finch, J. F. (1997). Competitiveness

mediates the link between personality and group performance. *Journal of Personality and Social Psychology, 73,* 1394–1408.

Graziano, W. G., Jensen-Campbell, L. A., & Finch, J. F. (1997). The self as a mediator between personality and adjustment. *Journal of Personality and Social Psychology, 73,* 392–404.

Graziano, W. G., Jensen-Campbell, L. A., & Hair, E. (1996). Perceiving interpersonal conflict and reacting to it: The case for agreeableness. *Journal of Personality and Social Psychology, 70,* 820–835.

Graziano, W. G., & Ward, D. (1992). Probing the Big Five in adolescence: Personality and adjustment during a developmental transition. *Journal of Personality, 60,* 425–440.

Graziano, W. G., & Waschull, S. B. (1995). Social development and self-monitoring. *Review of Personality and Social Psychology: Social Development, 15,* 233–260.

Halverson, C. F. (1988). Remembering your parents: Reflections on the retrospective method. *Journal of Personality, 56,* 435–444.

Halverson, C. F., & Post-Gordon, J. C. (1984). The measurement of open-field activity in preschool children. In E. Pollitt & P. Amante (Eds.), *Protein intake and activity* (pp. 185–206). New York: Alan R. Liss.

Halverson, C. F., & Waldrop, M. F. (1973). The relation of mechanically recorded activity levels to varieties of preschool play behavior. *Child Development, 44,* 371–378.

Halverson, C. F., & Waldrop, M. F. (1976). The relations between preschool activity and aspects of intellectual and social behavior at age 7½. *Developmental Psychology, 12,* 107–112.

Halverson, C., & Wampler, K. (1997). Family influences on personality development. In R. Hogan, J. Johnson, & S. Briggs (Eds.), *Handbook of personality psychology* (pp. 241–267). Orlando, FL: Academic Press.

Higgins, E. T., & Eccles Parsons, J. (1983). Social cognition and the social life of the child: Stages as subcultures. In E. T. Higgins, D. N. Ruble, & W. W. Hartup (Eds.), *Social cognition and social development: A sociocultural perspective* (pp. 15–62). New York: Cambridge University Press.

Hofstee, W. K. B. (1991). The concepts of personality and temperament. In J. Strelau & A. Angleitner (Eds.), *Exploration in temperament: International perspectives on theory and measurement* (pp. 177–190). New York: Plenum.

Ickes, W. (1982). A basic paradigm for study of personality, roles, and social behavior. In W. Ickes & E. S. Knowles (Eds.), *Personality, roles, and social behavior* (pp. 305–342). New York: Springer-Verlag.

Jensen-Campbell, L. A., Graziano, W. G., & Hair, E. C. (1996). Personality and relationships as moderators of interpersonal conflict in adolescence. *Merrill-Palmer Quarterly, 42,* 148–163.

John, O. P. (1990). The five-factor taxonomy: Dimensions of personality in the natural language and in questionnaires. In L. Pervin (Ed.), *Handbook of personality theory and research* (pp. 66–100). New York: Guilford Press.

Kenrick, D. T., & Stringfield, D. O. (1980). Personality traits and the eye of the beholder: Crossing some traditional philosophical boundaries in the search for consistencies in all of the people. *Psychological Review, 87,* 88–104.

Kernis, M. H., & Waschull, S. B. (1995). The interactive roles of stability and level of self-esteem: Research and theory. *Advances in Experimental Social Psychology, 27,* 93–141.

Lengua, L. J., West, S. G., & Sandler, I. N. (1998). Temperament as a predictor of symptomatology in children: Addressing contamination of measures. *Child Development, 69,* 164–181.

Lenney, E. (1991). Sex roles: The measurement of masculinity, femininity, and androgyny. In J. P. Robinson, P. R. Shaver, & L. S. Wrightsman (Eds.), *Measures of personality and social psychological attitudes* (Vol. 1, pp. 573–660). San Diego, CA: Academic Press.

Lerner, R. M., & Tubman, J. G. (1989). Conceptual issues in studying continuity and discontinuity in personality development across life. *Journal of Personality, 57,* 343–374.

Lewis, M. (1997). *Altering fate: Why the past does not predict the future.* New York: Guilford Press.

Madon, S., Jussim, L., & Eccles, J. (1997). In search of the powerful self-fulfilling prophecy. *Journal of Personality and Social Psychology, 72,* 791–809.

Martin, R. P. (1988). *The Temperament Assessment Battery for Children.* Brandon, VT: Clinical Psychology Publishing.

Martin, R. P., & Halverson, C. F. (1991). Mother–father agreement in temperament ratings: A preliminary investigation. In J. Strelau & A. Angleitner (Eds.), *Exploration in temperament: International perspectives on theory and measurement* (pp. 235–248). New York: Plenum.

Martin, R. P., Wisenbaker, J., & Huttunen, M. (1994). Review of factor analytic studies of temperament measures based on the Thomas-Chess structural model: Implications for the Big Five. In C. F. Halverson, D. Kohnstamm, & R. Martin (Eds.), *The developing structure of temperament and personality from infancy to adulthood* (pp. 157–172). Hillsdale, NJ: Erlbaum.

Matheny, A. P. (1989). Children's behavioral inhibition over age and across situations: Genetic similarity for a trait during change. *Journal of Personality, 57,* 343–374.

McCrae, R. R., & Costa, P. T. (1985). Updating Norman's "adequate taxonomy": Intelligence and personality dimensions in the natural language and in questionnaires. *Journal of Personality and Social Psychology, 49,* 710–721.

McCrae, R. R., & Costa, P. T. (1987). Validation of the five-factor model of personality across instruments and observers. *Journal of Personality and Social Psychology, 52,* 81–90.

McCrae, R. R., & Costa, P. T. (1997). Conceptions and correlates of openness to experience. In R. Hogan, J. Johnson, & S. Briggs (Eds.), *Handbook of personality psychology* (pp. 826–847). Orlando, FL: Academic Press.

McCrae, R. R., & John, O. J. (1992). An introduction to the five-factor model and its applications. *Journal of Personality, 60,* 175–216.

Miller, S. A. (1995). Parents' attributions for their children's behavior. *Child Development, 66,* 1557–1584.

Morris, L. W. (1979). *Extraversion and introversion: An interactional perspective.* New York: Wiley.

Musser, L., & Graziano, W. G. (1991). Behavioral confirmation in children's interactions with peers. *Basic and Applied Social Psychology, 12,* 441–456.

Olson, J. M., Roese, N. J., & Zanna, M. P. (1996). Expectancies. In E. T. Higgins & A. W. Kruglanski (Eds.), *Social psychology: Handbook of basic principles* (pp. 211–238). New York: Guilford Press.

Ozer, D. J., & Reise, S. P. (1994). Personality assessment. *Annual Review of Psychology, 45,* 357–388.

Panter, A. T., Tanaka, J. S., & Hoyle, R. H. (1994). Structural models for multimode designs in personality and temperament research. In C. F. Halverson, D. Kohnstamm, & R. Martin (Eds.), *Development of the structure of temperament and personality from infancy to adulthood* (pp. 111–138). Hillsdale, NJ: Erlbaum.

Plomin, R., DeFries, J. C., & Loehlin, J. C. (1977). Genotype–environment interaction and correlation in the analysis of human behavior. *Psychological Bulletin, 84,* 309–322.

Rickard, K., Graziano, W. G., & Forehand, R. (1984). Parental expectations and childhood deviancy in clinic-referred and non-clinic children. *Journal of Clinical Child Psychology, 13,* 179–186.

Rothbart, M. K. (1991). Temperament: A developmental framework. In J. Strelau & A. Angleitner (Eds.), *Exploration in temperament: International perspectives on theory and measurement* (pp. 61–74). New York: Plenum.

Rothbart, M. K., & Bates, J. E. (1998). Temperament. In W. Damon (Series Ed.) & N. Eisenberg (Vol. Ed.), *Handbook of child psychology: Vol. 3. Social, emotional, and personality development* (5th ed., pp. 105–176). New York: Wiley.

Sanson, A., & Rothbart, M. K. (1995). Child temperament and parenting. In M. Bornstein (Ed.), *Handbook of parenting: Vol. 4. Applied and practical parenting* (pp. 299–321). Hillsdale, NJ: Erlbaum.

Scarr, S., & McCartney, K. (1983). How people make their own environments: A theory of genotype → environment effects. *Child Development, 54,* 424–435.

Strelau, J. (1991). Renaissance in research on temperament: Where to. . .? In J. Strelau & A. Angleitner (Eds.), *Exploration in temperament: International perspectives on theory and measurement* (pp. 337–358). New York: Plenum.

Swann, W. B., Jr., & Ely, R. J. (1984). A battle of the wills: Self-verification versus behavioral confirmation. *Journal of Personality and Social Psychology, 46,* 1287–1302.

Tooby, J., & Cosmides, L. (1990). On the universality of human nature and the uniqueness of the individual: The role of genetics and adaptation. *Journal of Personality, 58,* 17–68.

Victor, J. B., & Halverson, C. F. (1975). Distractibility and hypersensitivity: Two behavioral factors in elementary school children. *Journal of Abnormal Child Psychology, 3,* 83–94.

von Eye, A. (1990). *Introduction to configural frequency analysis: The search for types and antitypes in cross-classifications.* New York: Cambridge University Press.

Wachs, T. D. (1994). Fit, context, and the transition between temperament and personality. In C. F. Halverson, D. Kohnstamm, & R. Martin (Eds.), *Development of the structure of temperament and personality from infancy to adulthood* (pp. 209–220). Hillsdale, NJ: Erlbaum.

Waddington, C. H. (1962). *New patterns in genetics and development.* New York: Columbia University Press.

Weiss, B., Dodge, K. A., Bates, J. E., & Pettit, G. S. (1992). Some consequences of early harsh punishment: Child aggression and maladaptive social information processing style. *Child Development, 63,* 1321–1335.

West, S. G., & Graziano, W. G. (1989). Long-term stability and change in personality: An introduction. *Journal of Personality, 57,* 175–194.

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